

Contrôle continu n° 1-S3

Durée : 30 min

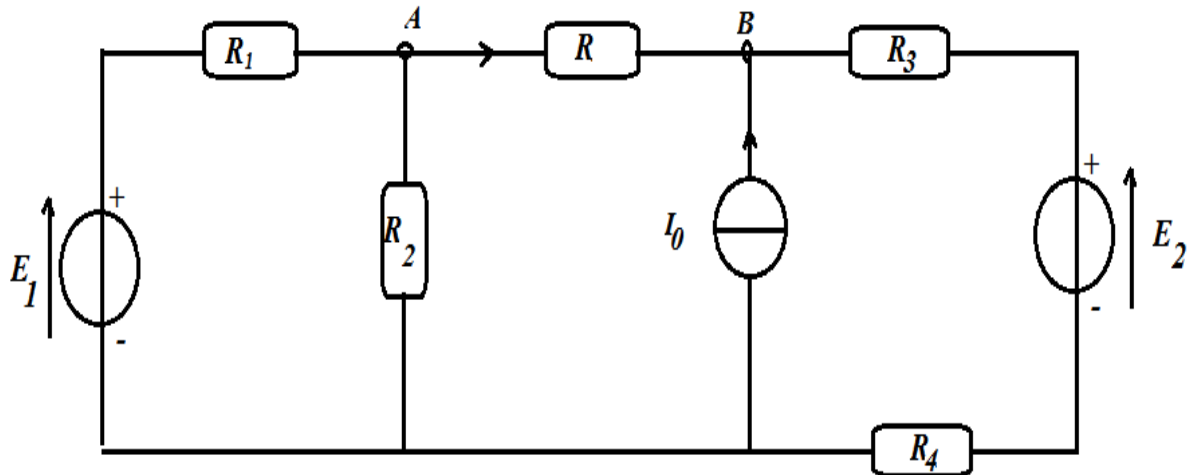
Exercice : 1

Soit le circuit ci-dessous :

On donne :

$E_1=50V$, $E_2=40V$, $I_0 = 8 A$, $R_1 = 10 \Omega$, $R_2 = 40 \Omega$, $R_3 = 10 \Omega$, $R_4 = 10\Omega$ et $R=7 \Omega$.

La résistance R étant le dipôle récepteur du circuit. Utilisons la transformation Norton – Thévenin pour calculer le courant qui circule dans la résistance R .



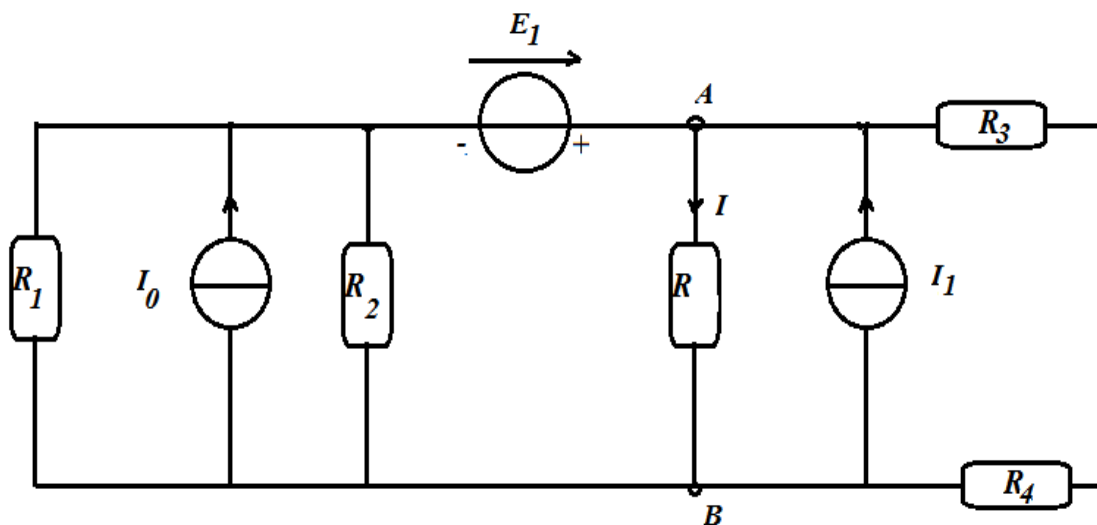
Exercice : 2

Soit le circuit ci-dessous :

On donne :

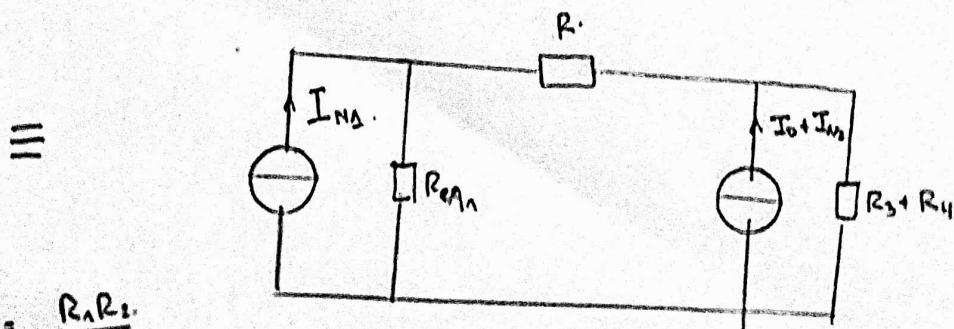
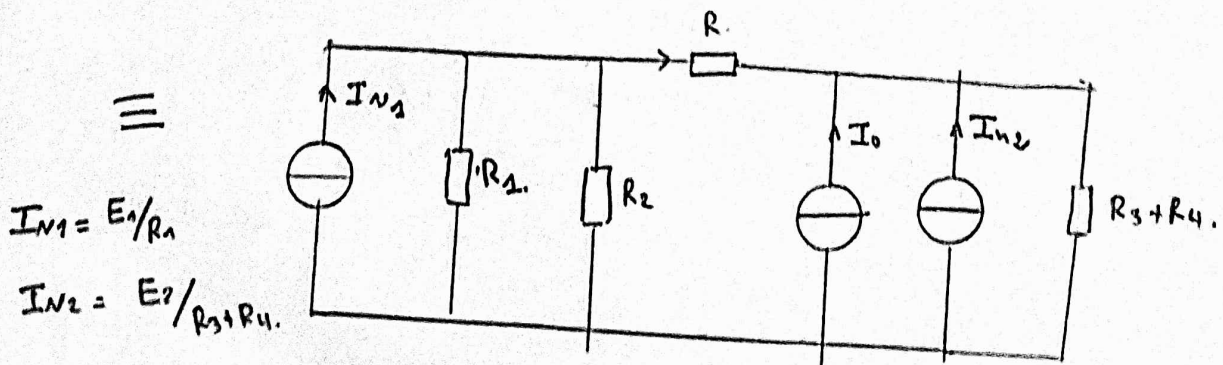
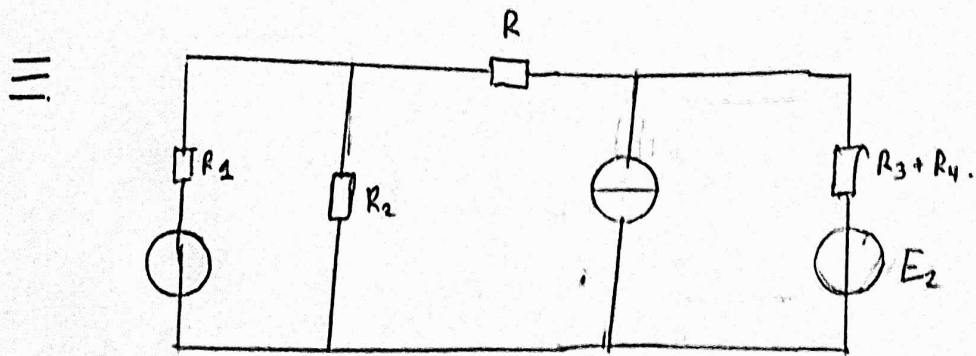
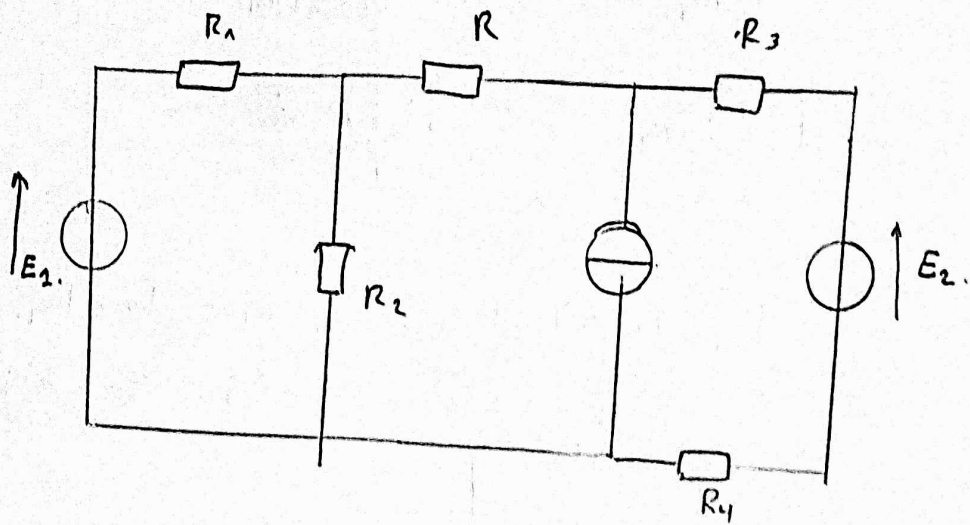
$E_1=5V$, $I_0=5A$, $I_1 = 3 A$, $R_1 = 6 \Omega$, $R_2 = 3 \Omega$, $R_3 = 1\Omega$, $R_4 = 4\Omega$ et $R=7 \Omega$.

La résistance R étant le dipôle récepteur du circuit. Utilisons la transformation Norton – Thévenin pour calculer le courant qui circule dans la résistance R .

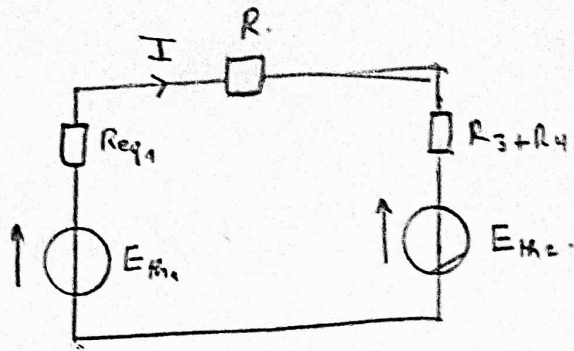


Contrôle continu n°1.

Exercice 1.

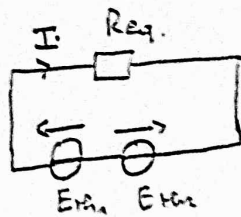


$$R_{eq1} = \frac{R_1 R_2}{R_1 + R_2}$$



$$R_{eq} = R_{eq1} + R + R_3 + R_4 \quad , \quad E_{K1} = I_{N1} \times R_{eq1}$$

$$E_{K2} = (I_{N2} + I_0)(R_3 + R_4)$$



$$I = \frac{E_{K1} - E_{K2}}{R_{eq}}$$

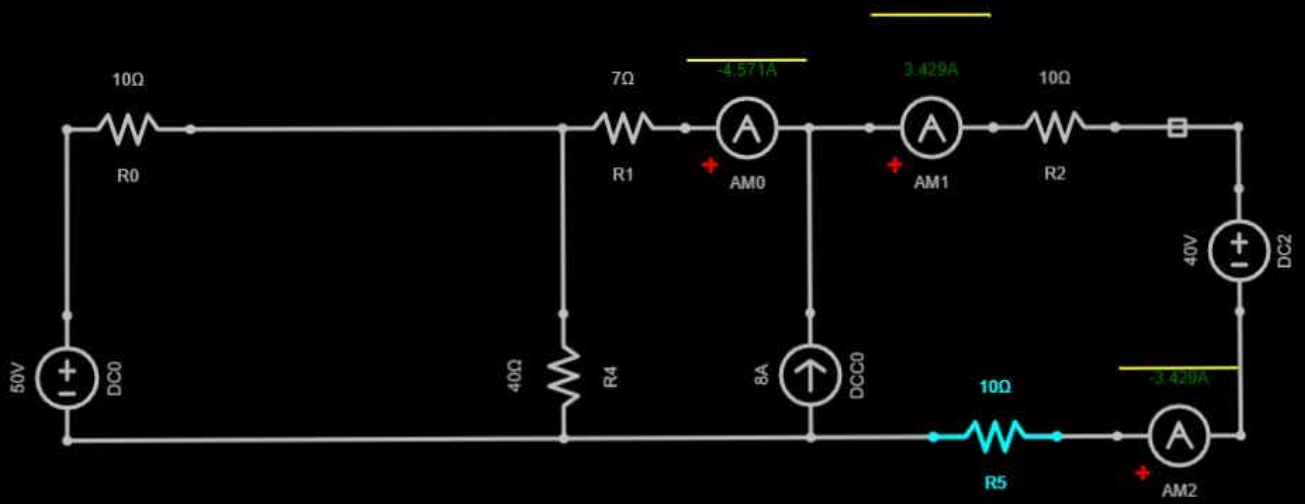
$$= \frac{I_{N1} \times R_{eq1} - (I_{N2} + I_0)(R_3 + R_4)}{R_{eq1} + R + R_3 + R_4}$$

$$= \frac{\frac{E_1}{R_1} \times \frac{R_1 R_2}{R_1 + R_2} - \left(\frac{E_2}{R_3 + R_4} + I_0 \right) (R_3 + R_4)}{\frac{R_1 R_2}{R_1 + R_2} + R + R_3 + R_4}$$

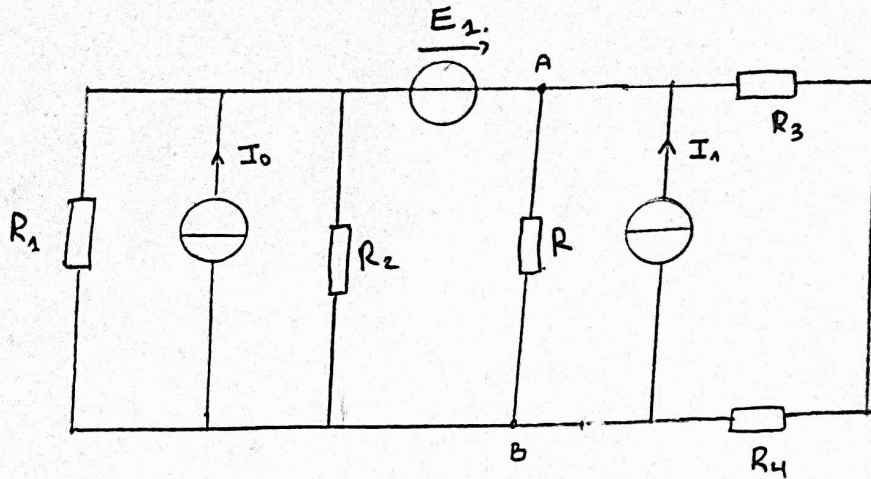
$$= \frac{\frac{E_1 R_2}{R_1 + R_2} - (E_2 + I_0(R_3 + R_4))}{\frac{R_1 R_2}{R_1 + R_2} + R + R_3 + R_4}$$

$$= \frac{\frac{50 \times 40}{10 + 40} - (40 + 8(10 + 10))}{\frac{10 \times 40}{10 + 40} + 7 + 10 + 10}$$

$$I = -4,57 \text{ A}$$



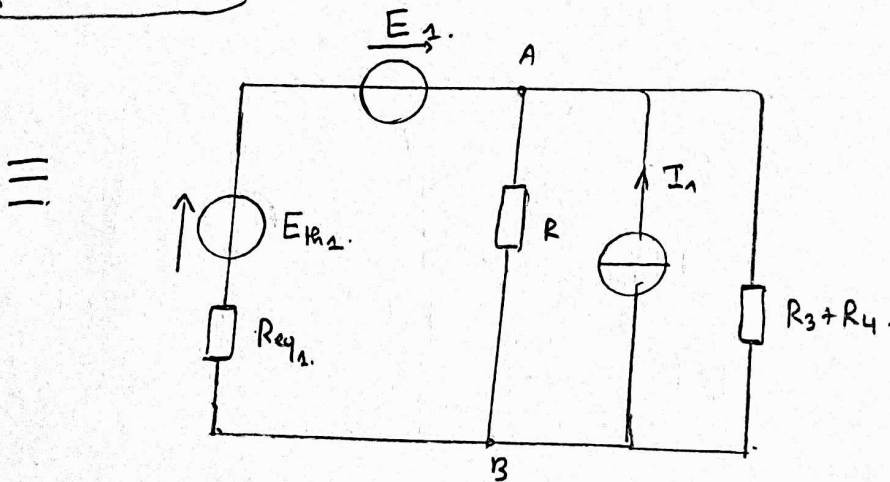
Exercice 2:



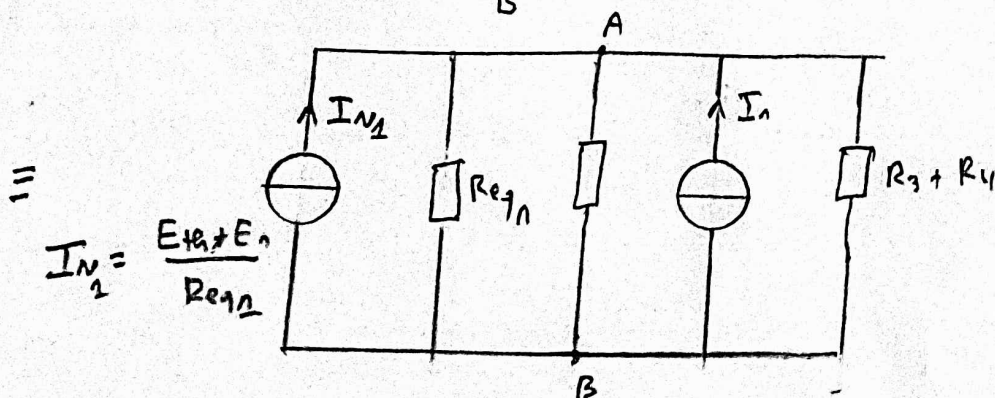
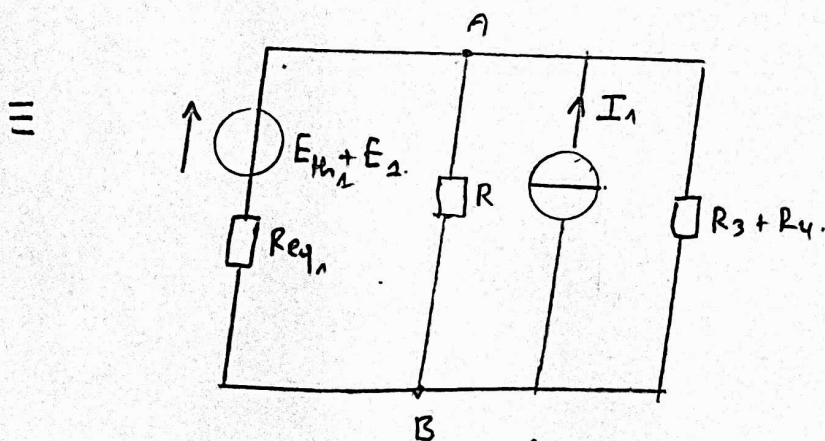
$$R_1 // R_2 \quad R_{eq1} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq1} = \frac{6 \times 3}{6 + 3} = 2 \Omega$$

R_3 en Serie avec R_4



$$E_{th1} = R_{eq1} \times I_0 = 2 \times 5 = 10 \text{ V} = E_{th1}$$



$$I_{N1} = \frac{E_{th1} + E_2}{R_{eq1}}$$

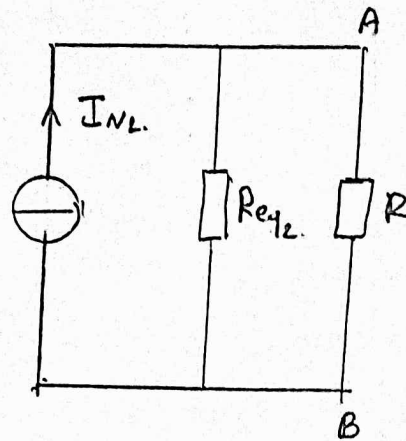
$$I_{N1} = \frac{10 + 5}{2} = 7.5 \text{ A}$$

$$R_{eq1} \parallel (R_3 + R_4)$$

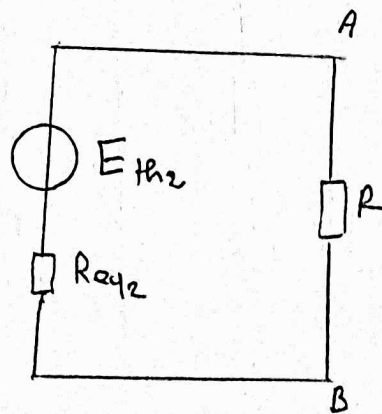
$$R_{eq2} = \frac{R_{eq1} \times (R_3 + R_4)}{R_{eq1} + R_3 + R_4} = \frac{2 \times (1 + 4)}{2 + 1 + 4} = \frac{10}{7} \Omega$$

$$I_{N2} = I_{N1} + I_{\Delta} = 7.5 + 3 = 10.5 \text{ A}$$

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$$\begin{aligned} E_{th2} &= I_{N2} \times R_{eq2} \\ &= 10.5 \times \frac{10}{7} \\ &= 15 \text{ V} \end{aligned}$$

$$I = \frac{E_{th2}}{R_{eq2} + R} = \frac{15}{\frac{10}{7} + 7}$$

$$I = 1.78 \text{ A}$$

